Campsite Management Systems

A Comprehensive Web Application Proposal

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Introduction

The Service Industry Context

The hospitality sector increasingly relies on tech solutions for efficiency. However, campsite management presents unique logistical challenges such as specific site allocation and equipment stock monitoring.

The Proposal: A comprehensive Campsite Management System (CMS) built as a web application using **React 18** and **TypeScript** to streamline operations for administrators, staff, and guests.

Problem Statement & Motivation

Many campsites still rely on fragmented tools or manual processes, leading to critical operational bottlenecks.

Operational Inefficiency

Manual booking and inventory tracking processes are highly prone to human error and lack real-time coordination.

Poor Visualization

Staff and customers lack visual tools to view or select specific campsite locations, leading to confusion.

Limited UX

Customers struggle with disjointed processes for viewing, managing, and paying for their outdoor stays.

Research Objectives

- **Role-Based Access:** Implement secure controls for Administrators, Managers, Staff, and Customers.
- **Interactive Map Editor:** Build a visual tool using Fabric.js for campsite layout design and site browsing.
- **Booking Workflow:** Develop a system supporting both staff-managed and customer-initiated reservations.
- **Equipment Rentals:** Create a rental system that seamlessly integrates with site bookings.

Project Scope & Limitations

Scope

- Single Page Application (SPA) using React 18, Vite, and TypeScript.
- **Key Modules:** Authentication, Booking Management, Interactive Maps, Equipment Rentals, and Analytics.

Limitations

- **Payments:** Uses mock mode (Stripe integration planned for future).
- **Authentication:** Currently uses mock credentials due to the absence of a live backend in this phase.

Literature Review

SYSTEM	STRENGTHS	WEAKNESSES
ResNexus	Online booking, marketing integration, and built-in Guest Portal/POS.	Not specific to campgrounds; OTA model disadvantages smaller sites.
RMS Cloud	Centralized operations for large parks; strong group reservation handling.	High complexity and steep learning curve; pricing varies by size.
WebRezPro	Flexible; supports packaged bookings (tours/activities).	Lack of accessible APIs, language limitations, non-transparent pricing.

Critical Analysis: *Existing solutions suffer from complexity, generic hotel-centric designs, and limited customization.*

Proposed Solution

Bridging the Gap: The proposed system addresses market shortcomings by focusing specifically on campsite logistics.

- **Streamlined Dashboard:** A clean, role-based interface built with React 18.
- **Visual Selection:** Interactive map editor allows users to choose specific pitches.
- **Modularity:** Designed for future integrations (payments, IoT).
- **Real-time Availability:** Direct focus on site and equipment logistics.

Methodology & Tech Stack

Agile development model with sprints to accommodate emerging requirements.

Frontend Core

React 18 with Vite for rapid development. TypeScript ensures type safety across the application.

State & Logic

Zustand and React Query for managing complex UI and server states efficiently.

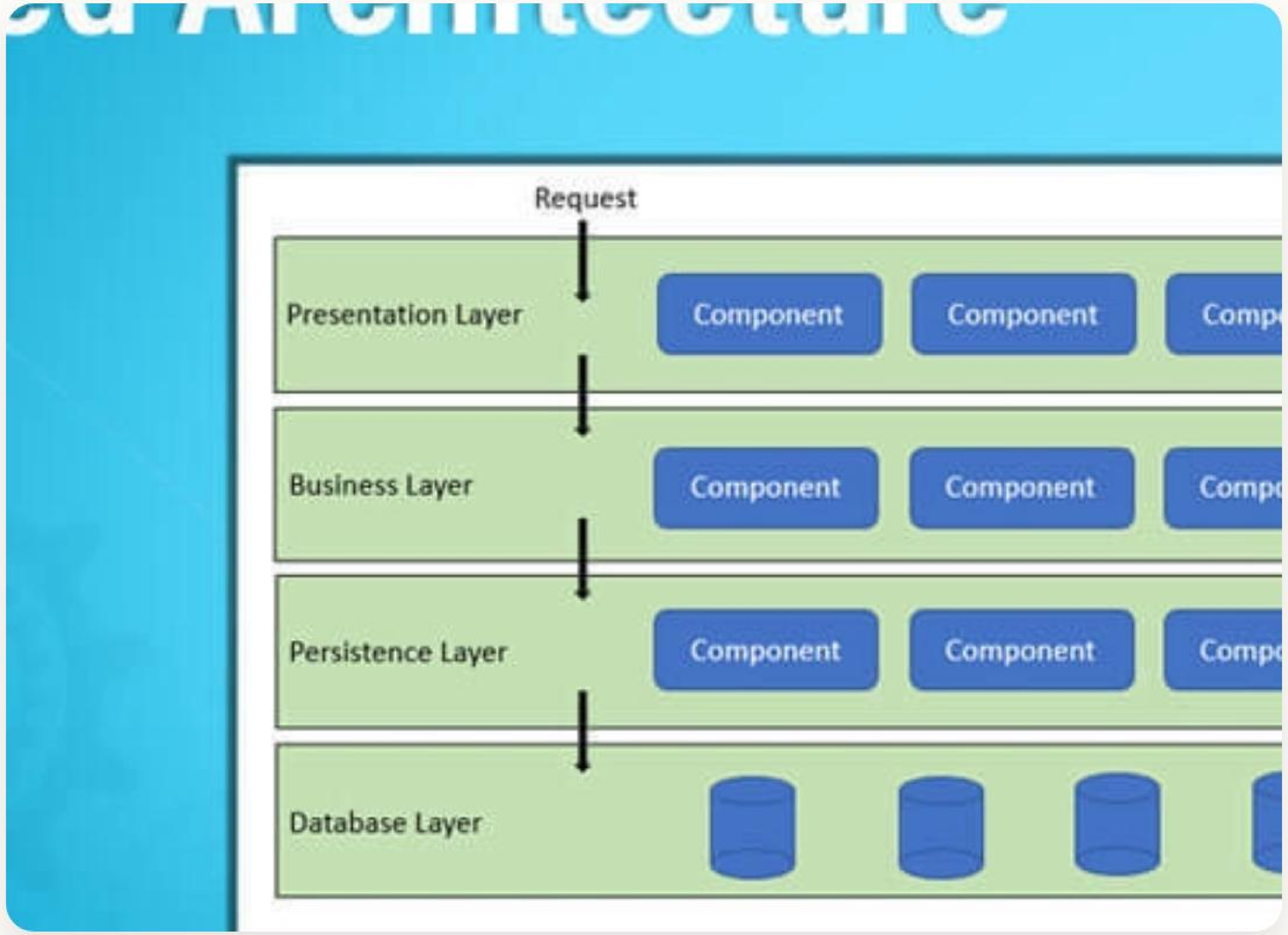
Visualization

Fabric.js powers the interactive map editor, enabling drag-and-drop site layout design.

System Architecture

The system follows a modular four-layer web architecture:

- **UI Layer:** React 18 + Tailwind CSS + Framer Motion.
- **Business Logic:** Booking validation, rental computation, RBAC (Zustand).
- **Integration Layer:** Manages external services (mock/live).
- **Data Layer:** Structured JSON (Project I) migrating to PostgreSQL (Project II).



Preliminary Work

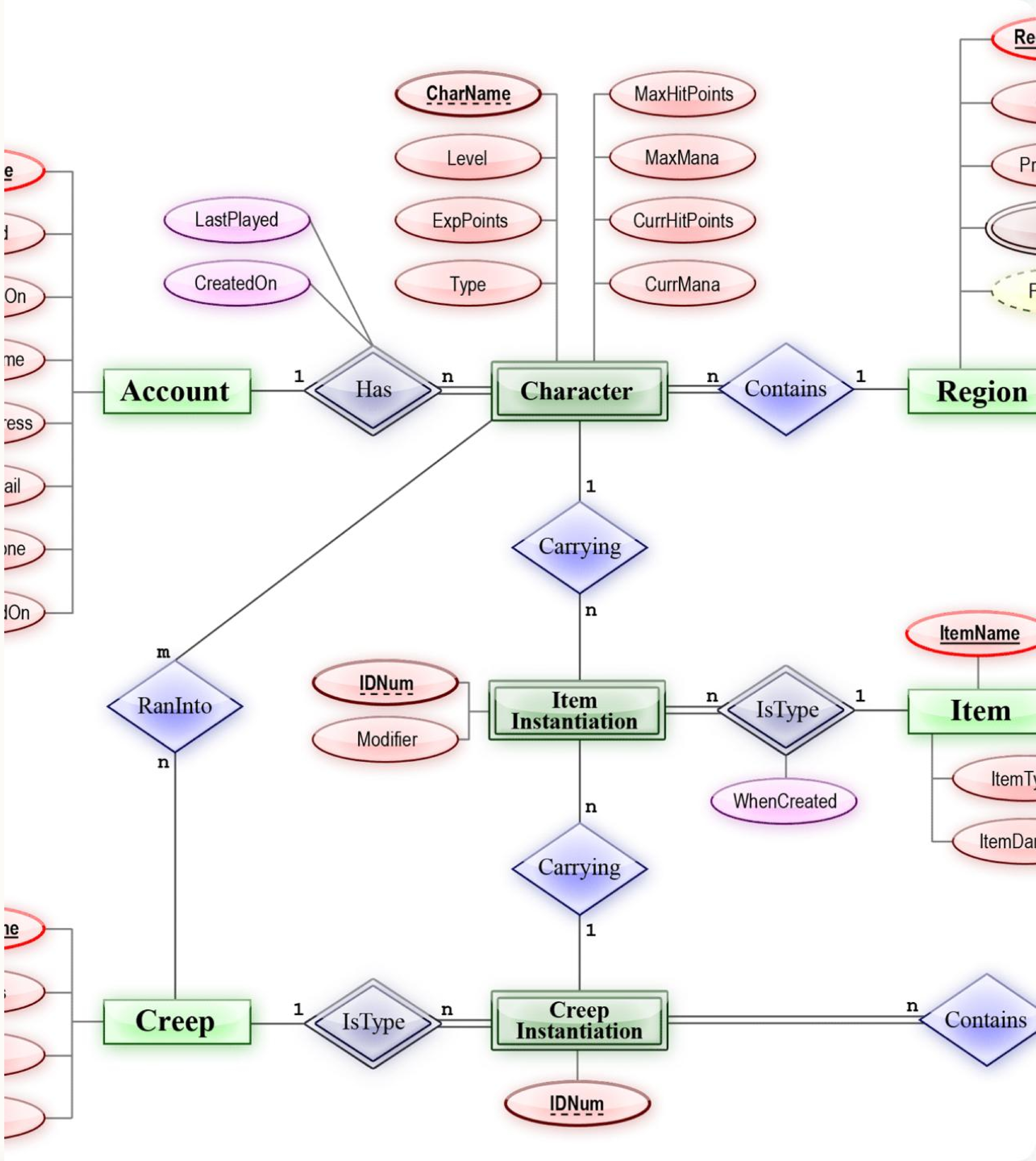
Project I Achievements, Data Design, and Feasibility Analysis

Data Schema Design

Although currently running on mock data, a robust database schema has been designed for the backend migration.

Key Entity: User

- Central to the system.
- Connects to Bookings, Notifications, and Roles.
- Enforces Role-Based Access Control (ADMIN, MANAGER, STAFF, CUSTOMER).



Development Timeline

Phase 1

Nov 3 - Dec 12

Core Auth, Site Mgmt, Booking Logic

Frontend

React 18 & Fabric.js

Interactive Map Editor Prototype

Backend

Project II

Node.js, Express, PostgreSQL Migration

Mobile App

Future Work

React Native for On-site Check-in

Conclusion

Summary & Future Path

The project has successfully delivered a high-fidelity prototype with an intuitive map editor and robust state management.

Next Steps:

- Migrate to Node.js/PostgreSQL backend for persistent storage.
- Implement secure authentication and concurrency control.
- Develop React Native mobile app for offline access.

Q & A

Thank you for your attention.

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Image Sources

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Source: glitzcamp.com

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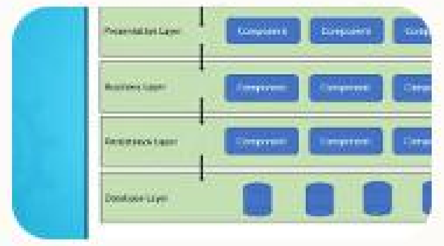

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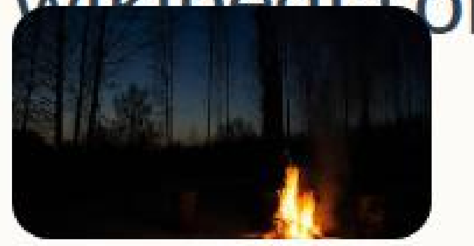

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